NEXT GEN
SOFTWARE HUB

CYBER SECURITY INTERNSHIP

WEEK 02
TASK 02 REPORT

Networking Fundamentals

Understanding the OSI/TCP-IP Model,
IP, Subnetting, DNS, and DHCP

TOPICS

OSI/TCP-IP Model, IP,
Subnetting, DNS, DHCP

TOOLS

Wireshark

Packet Tracer

RESOURCES

Computer Networking
Full Course

Wireshark Tutorial

TASK

Capture a local packet using
Wireshark & analyze HTTP/HTTPS.

Week 2: Networking Fundamentals

PLANS :

Phase 1: Foundations (Weeks 1–4)



PREPARED BY

Selo David Sethibo
Cyber Security Intern

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1. INTRODUCTION

Computer networking is the process of connecting computers, servers, network devices, applications, and other systems so that they can communicate and exchange information. Networks allow users to access websites, communicate with servers, share resources, use applications, and access network services.

Networking is an important foundation of cybersecurity because most modern systems depend on network communication. When a user accesses a website, logs into an application, connects to a server, or transfers information, network traffic is generated between different systems.

Cybersecurity professionals therefore need to understand how network communication works and how to identify, analyze, and monitor network traffic.

This task focuses on networking fundamentals, including the OSI Model, TCP/IP Model, IP addressing, subnetting, DNS, DHCP, TCP, UDP, network devices, firewalls, VPNs, network packets, Nmap, and Wireshark.

The practical component was performed in a controlled virtual laboratory using Kali Linux as the analysis machine and DC-1 from VulnHub as the intentionally vulnerable target machine.

The practical involved discovering the target machine using Netdiscover, identifying its IP address, scanning the target using Nmap, identifying available network services, accessing the HTTP service using Firefox, and capturing and analyzing the resulting network traffic using Wireshark.

2. OBJECTIVES

The objectives of this task were:

- To understand the fundamentals of computer networking.
- To understand the OSI Model.
- To understand the TCP/IP Model.
- To understand IP addresses.
- To understand subnetting.
- To understand TCP and UDP.
- To understand DNS and DHCP.
- To understand common network devices.
- To understand network packets.
- To understand the purpose of firewalls.
- To understand the purpose of VPNs.
- To understand basic network security.
- To understand the purpose of Nmap.
- To understand the purpose of Wireshark.
- To discover a target machine using Netdiscover.
- To identify the IP address of the target.
- To scan the target using Nmap.
- To identify open ports and services.
- To access an HTTP service.
- To capture HTTP network traffic using Wireshark.
- To analyze captured HTTP packets.
- To connect theoretical networking concepts with practical network traffic analysis.

3. INTRODUCTION TO COMPUTER NETWORKING

A computer network is a group of connected devices that communicate with one another and exchange information.

Networked devices can include:

- Computers
- Laptops
- Servers
- Routers
- Switches
- Firewalls
- Printers
- Smartphones
- IoT devices
- Virtual machines

Networks allow devices to communicate and share resources.

For example, when a user enters a website address into a browser, the computer communicates with a server. The browser sends a request to the server, and the server responds with the requested information.

Network communication relies on protocols and addressing systems that allow information to be delivered to the correct destination.

Important networking concepts include:

- IP addresses
- MAC addresses
- Ports
- Protocols
- Packets
- Network devices
- Network services

Understanding these concepts is important in cybersecurity because network traffic can provide information about systems, services, users, and communication patterns.

4. ENTERPRISE NETWORKING

Enterprise networking refers to the network infrastructure used by organizations to connect employees, computers, servers, offices, applications, data centers, cloud environments, and other resources.

An enterprise network may contain:

- Local Area Networks (LANs)
- Wide Area Networks (WANs)
- Branch offices
- Data centers
- Cloud environments
- Servers

- Employee computers
- Network switches
- Routers
- Firewalls
- VPN infrastructure
- IoT devices

Enterprise networks allow different systems and users to communicate and access organizational resources.

Modern enterprise environments may also use technologies such as cloud computing, VPNs, SD-WAN, network segmentation, and secure remote access.

From a cybersecurity perspective, understanding enterprise networking is important because security controls must protect communication between different systems and network segments.

5. NETWORK DEVICES

Different network devices perform different functions.

5.1 Routers

A router connects different networks.

It forwards packets between networks using IP addresses and routing information.

For example, a home router can connect a private local network to the Internet.

5.2 Switches

A switch connects devices within a local network.

Switches commonly operate at the Data Link Layer and use MAC addresses to forward Ethernet frames.

5.3 Gateways

A gateway provides communication between different networks or systems.

A default gateway allows devices on a local network to communicate with destinations outside their local network.

5.4 Firewalls

A firewall monitors and controls network traffic according to security rules.

A firewall can allow legitimate traffic and block unauthorized connections.

5.5 Load Balancers

A load balancer distributes network or application requests across multiple servers.

This can improve performance, reliability, and availability.

5.5 VPN Servers

A VPN server allows authorized users to establish secure connections to private network resources over another network, such as the Internet.

6. OSI MODEL

The Open Systems Interconnection (OSI) Model is a conceptual model used to explain how network communication occurs.

It consists of seven layers:

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

6.1 Physical Layer

The Physical Layer deals with the transmission of raw bits through physical or wireless media.

Examples include:

- Ethernet cables
- Fiber-optic cables
- Wireless signals
- Network connectors

6.2 Data Link Layer

The Data Link Layer handles communication between devices on the same local network.

It uses MAC addresses and Ethernet frames.

Switches commonly operate at this layer.

6.3 Network Layer

The Network Layer handles logical addressing and routing.

IP operates at this layer.

In the practical environment, the Kali Linux machine had:

Source IP: 192.168.94.128

and DC-1 had:

Destination IP: 192.168.94.134

Wireshark displayed these addresses when the communication was captured.

6.4 Transport Layer

The Transport Layer provides communication between applications.

The two major transport protocols are:

- TCP
- UDP

TCP and UDP use port numbers to identify network services.

During the practical, HTTP communication used TCP.

The selected HTTP packet showed:

Source port: 57626

Destination port: 80

6.5 Session Layer

The Session Layer is responsible for establishing, maintaining, and terminating communication sessions between applications.

6.6 Presentation Layer

The Presentation Layer deals with how data is represented.

Functions associated with this layer include:

- Data formatting
- Encoding
- Encryption
- Compression

6.7 Application Layer

The Application Layer provides network services that applications use.

Examples include:

- HTTP
- HTTPS
- DNS
- FTP
- SSH

The practical demonstrated HTTP communication between Kali Linux and the DC-1 web server.

7. TCP/IP MODEL

The TCP/IP model describes the protocols and layers used by modern computer networks and the Internet.

The four commonly described layers are:

1. Application
2. Transport
3. Internet
4. Network Access

Examples include:

- Application: HTTP, HTTPS, DNS, SSH
- Transport: TCP, UDP
- Internet: IP
- Network Access: Ethernet and other network technologies

The TCP/IP model is important because it provides a practical framework for understanding how network communication occurs.

8. IP ADDRESSES AND SUBNETTING

Internet Protocol (IP) provides addressing and routing mechanisms for network communication.

An IP address identifies a network interface.

During the practical, the systems used the following addresses:

- Kali Linux: 192.168.94.128
- DC-1: 192.168.94.134

Both systems were located within the same private network.

Subnetting

Subnetting is the process of dividing a network into smaller logical networks.

A common example is: 192.168.94.128/24

A /**24** subnet corresponds to: **255.255.255.0**

Subnetting allows organizations to divide networks into manageable sections and can be useful for network organization, performance, and security segmentation.

For example, organizations may separate:

- User devices
- Servers
- Guest devices
- Security systems
- IoT devices

into different network segments.

9. TCP AND UDP

9.1 Transmission Control Protocol

TCP is a connection-oriented transport protocol.

It provides:

- ❖ Reliable delivery
- ❖ Ordered data
- ❖ Acknowledgements
- ❖ Retransmission
- ❖ Connection establishment

In the practical, HTTP traffic was transported using TCP.

The captured packet showed:

- Source port: 57626
- Destination port: 80

Therefore:

Kali Linux



DC-1

9.2 User Datagram Protocol

UDP is a connectionless transport protocol.

Unlike TCP, UDP does not establish a connection before sending data.

UDP does not guarantee:

- ❖ Delivery
- ❖ Ordering
- ❖ Retransmission

UDP has lower overhead than TCP and is useful for applications where speed is more important than guaranteed delivery.

Examples include DNS queries and some real-time communication.

9.3 TCP vs UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable delivery	No guaranteed delivery
Uses acknowledgements	No acknowledgements
Retransmits lost data	Does not provide retransmission
Higher overhead	Lower overhead
Example: HTTP	Example: DNS

10. DNS

Domain Name System (DNS) translates human-readable domain names into IP addresses.

For example:

www.example.com

can be resolved to an IP address.

Without DNS, users would generally need to remember IP addresses instead of domain names.

DNS is important for cybersecurity because DNS traffic can also provide useful information during network monitoring and security investigations.

For example, security analysts can investigate unusual DNS requests to identify potentially suspicious communication.

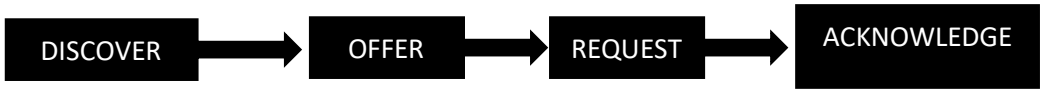
11. DHCP

Dynamic Host Configuration Protocol (DHCP) automatically provides network configuration information to devices.

DHCP can provide:

- IP address
- Subnet mask
- Default gateway
- DNS server

The basic DHCP process is commonly described using DORA:



DHCP reduces the need to manually configure IP addresses on every device.

12. NETWORK PACKETS

A network packet is a unit of information transmitted across a network.

A packet may contain information such as:

- Source IP
- Destination IP
- Protocol
- Source port
- Destination port
- Packet length
- Application data

Wireshark allows network professionals to inspect these details.

During the practical, a captured HTTP packet contained information from multiple networking layers:



This demonstrated how different protocols work together during network communication.

13. FIREWALLS

A firewall is a security mechanism that monitors and controls network traffic according to predefined rules.

Rules can consider:

- Source IP
- Destination IP
- Port
- Protocol
- Connection information

For example, a firewall could allow HTTPS traffic on port 443 while blocking unauthorized connections.

Firewalls are important because they help reduce the number of services and connections accessible to unauthorized users.

However, firewalls should be combined with other security controls such as:

- Authentication
- Access control
- Encryption
- Network monitoring
- Network segmentation
- Endpoint security

14. VIRTUAL PRIVATE NETWORKS

A Virtual Private Network (VPN) provides a secure connection between a user and a VPN endpoint.

VPNs are commonly used to allow authorized users to access private organizational resources remotely.

VPN technology can provide encryption that protects network communication from unauthorized observation while traffic travels across an untrusted network.

VPNs are therefore an important technology in remote access and enterprise network security.

15. NETWORK SECURITY

Network security involves protecting networks, devices, systems, communication, and information against unauthorized access and attacks.

Network security controls can include:

- Firewalls
- VPNs
- Authentication
- Access control

- ❖ Encryption
- ❖ Network segmentation
- ❖ Intrusion detection
- ❖ Intrusion prevention
- ❖ Network monitoring

Tools such as Nmap and Wireshark provide different types of network visibility.

- ➢ Nmap can identify hosts, open ports, and services.
- ➢ Wireshark can provide detailed visibility into network packets and protocols.

16. NMAP

Nmap stands for Network Mapper.

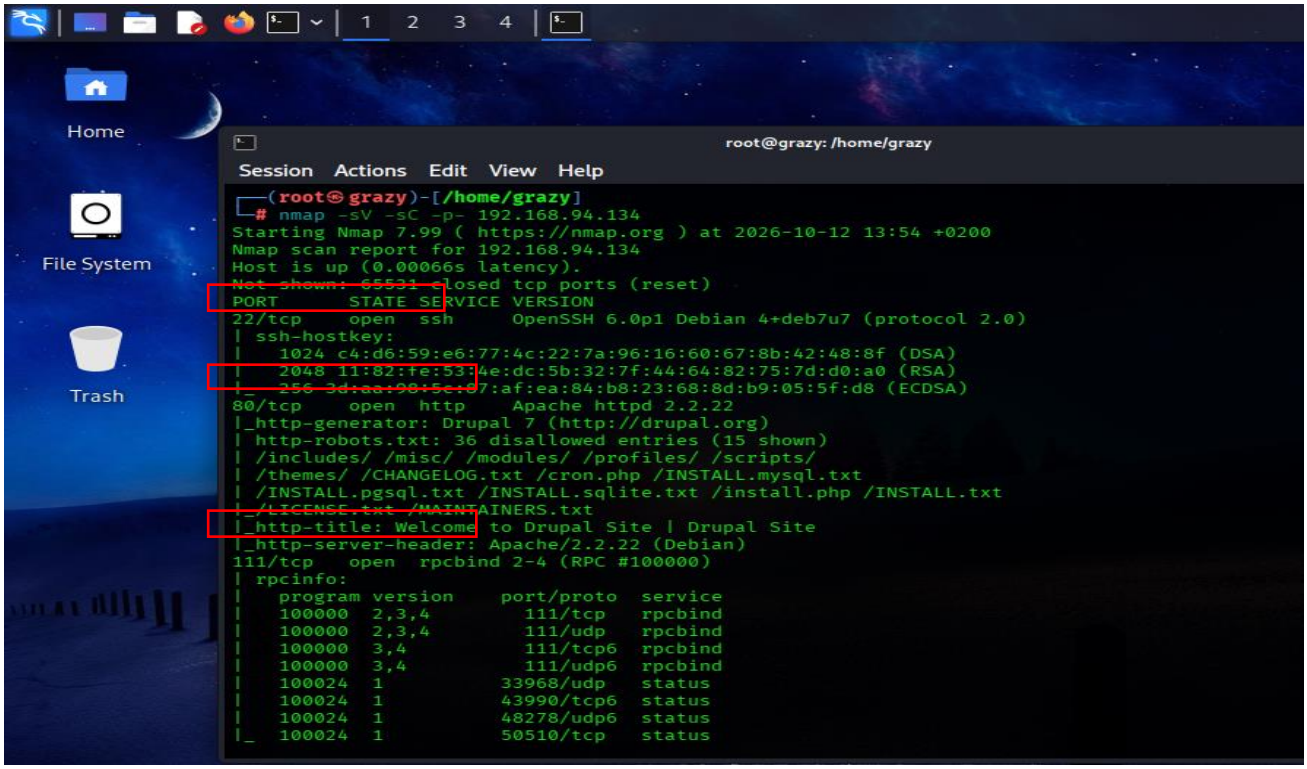
It is a network scanning tool used to:

- ❖ Discover hosts
- ❖ Identify open ports
- ❖ Identify services
- ❖ Identify service versions
- ❖ Gather information about a target system

Nmap can be useful to cybersecurity professionals when assessing authorized systems and identifying exposed services.

In this practical, Nmap was used against the intentionally vulnerable DC-1 machine from VulnHub.

The scan identified:



17. WIRESHARK

Wireshark is a network protocol analyzer used to capture and inspect network traffic.

It can display information such as:

- Source IP
- Destination IP
- Protocol
- Source port
- Destination port
- Packet length
- TCP information
- HTTP information
- TLS information

Wireshark can be used for:

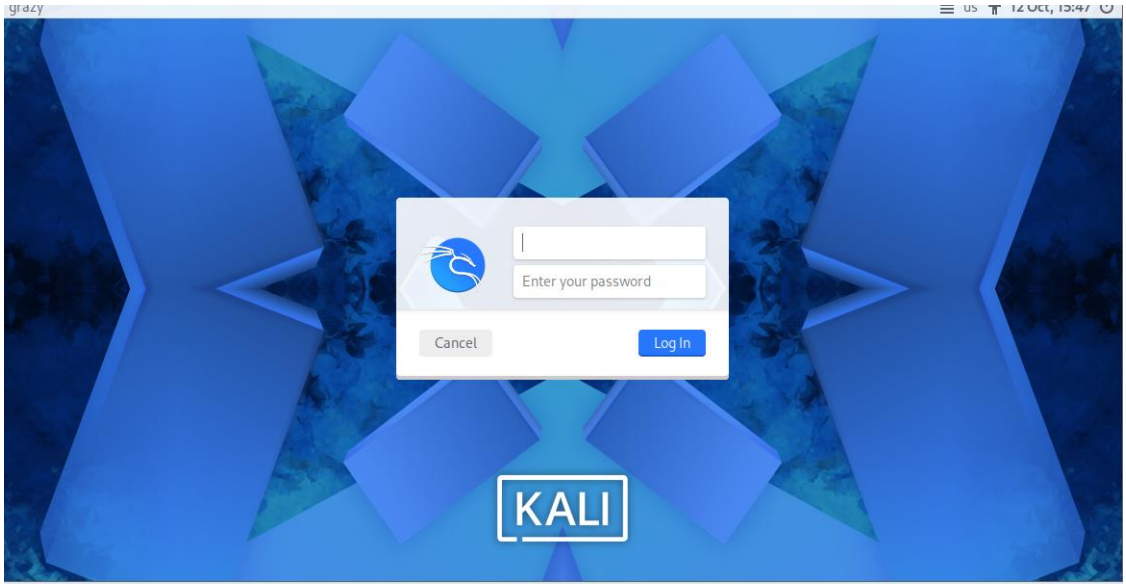
- Network troubleshooting
- Protocol analysis
- Security investigations
- Traffic analysis
- Incident response
- Learning networking concepts

In this practical, Wireshark was used to capture traffic generated when Kali Linux accessed the DC-1 web server.

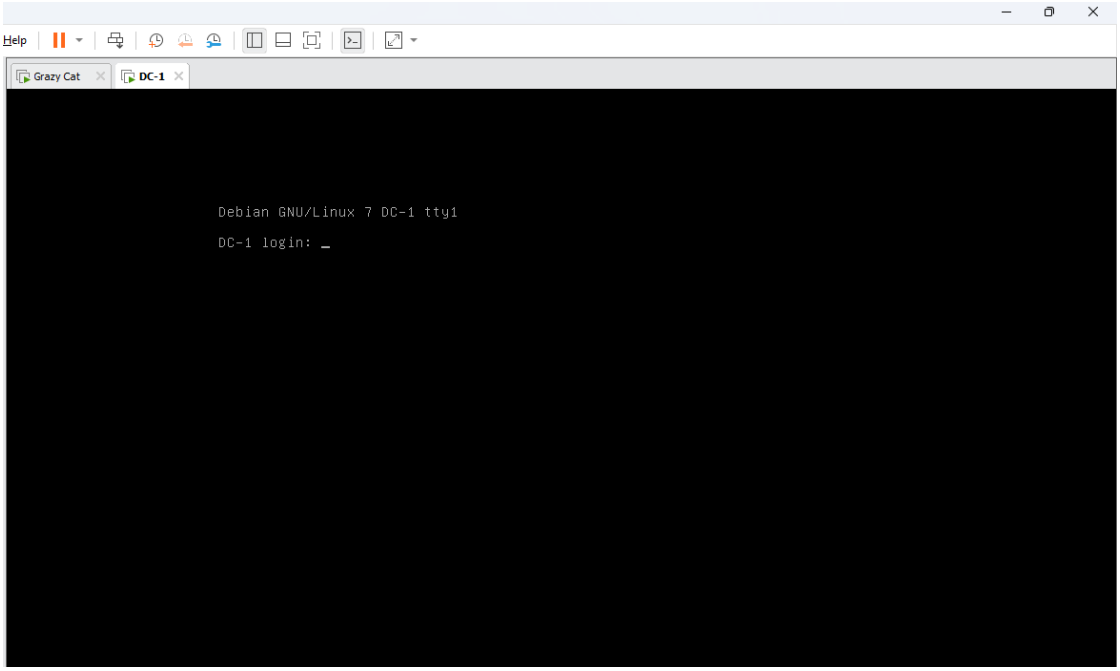
18. PRACTICAL ENVIRONMENT

The practical was performed in a controlled virtual laboratory.

Analysis machine: Kali Linux



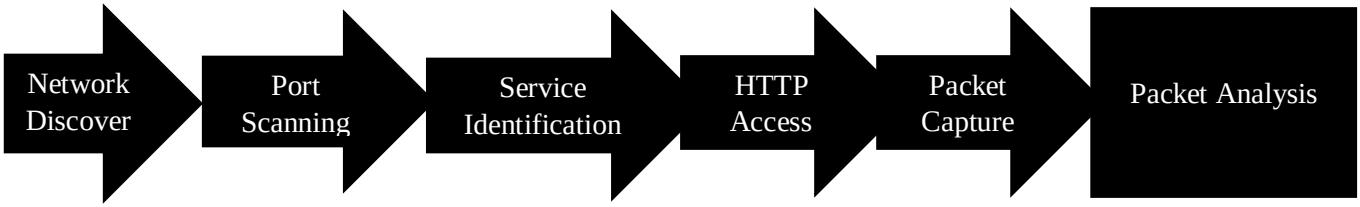
Target machine: DC-1 from VulnHub



Tools

- Netdiscover
- Nmap
- Wireshark
- Firefox

The practical followed a controlled network-analysis process:



19. PRACTICAL METHODOLOGY

The practical was completed using the following process:

1. Started the Kali Linux virtual machine.
2. Started the DC-1 virtual machine.
3. Used Netdiscover to identify devices on the local network.
4. Identified DC-1 at 192.168.94.134.
5. Used Nmap to scan the target.
6. Identified open ports and services.
7. Identified HTTP running on port 80.
8. Accessed the DC-1 web server using Firefox.
9. Started Wireshark and selected the appropriate network interface.
10. Captured network traffic generated by the web request.

11. Identified HTTP packets.
12. Selected a relevant HTTP packet for detailed analysis.
13. Examined the IP addresses and TCP ports.
14. Examined the HTTP request.
15. Identified the requested resource.
16. Connected the practical results to networking and cybersecurity concepts.

20. NETWORK DISCOVERY USING NETDISCOVER

The first practical step was discovering devices on the local network.

The following command was used:

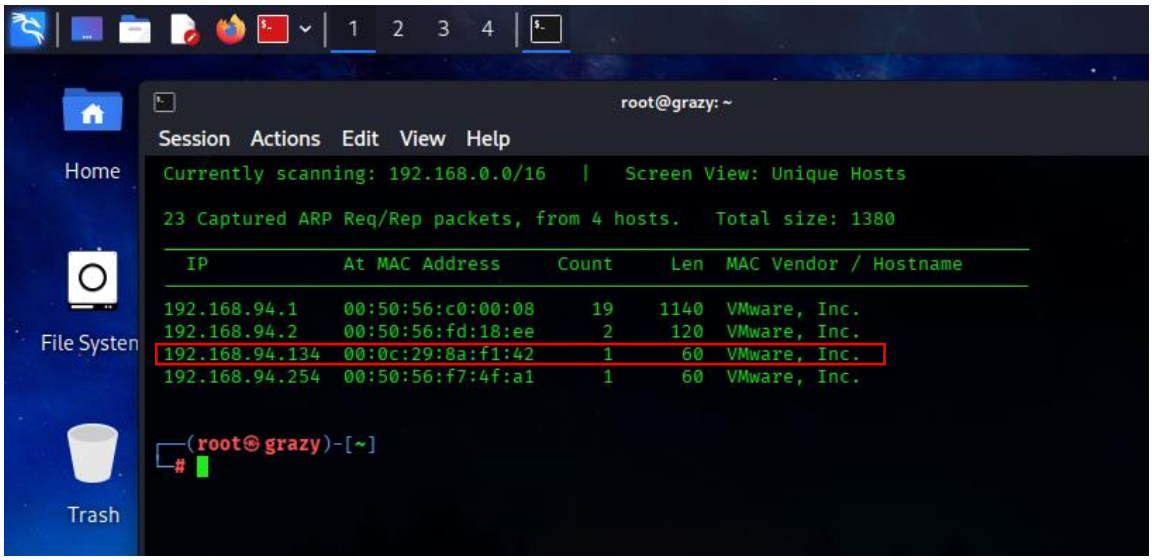
netdiscover

The results showed several devices, including:

- 192.168.94.1
- 192.168.94.2
- 192.168.94.134
- 192.168.94.254

The DC-1 target was identified at: 192.168.94.134

This address was then used during the subsequent Nmap scan.



Network discovery using Netdiscover

What this screenshot proves

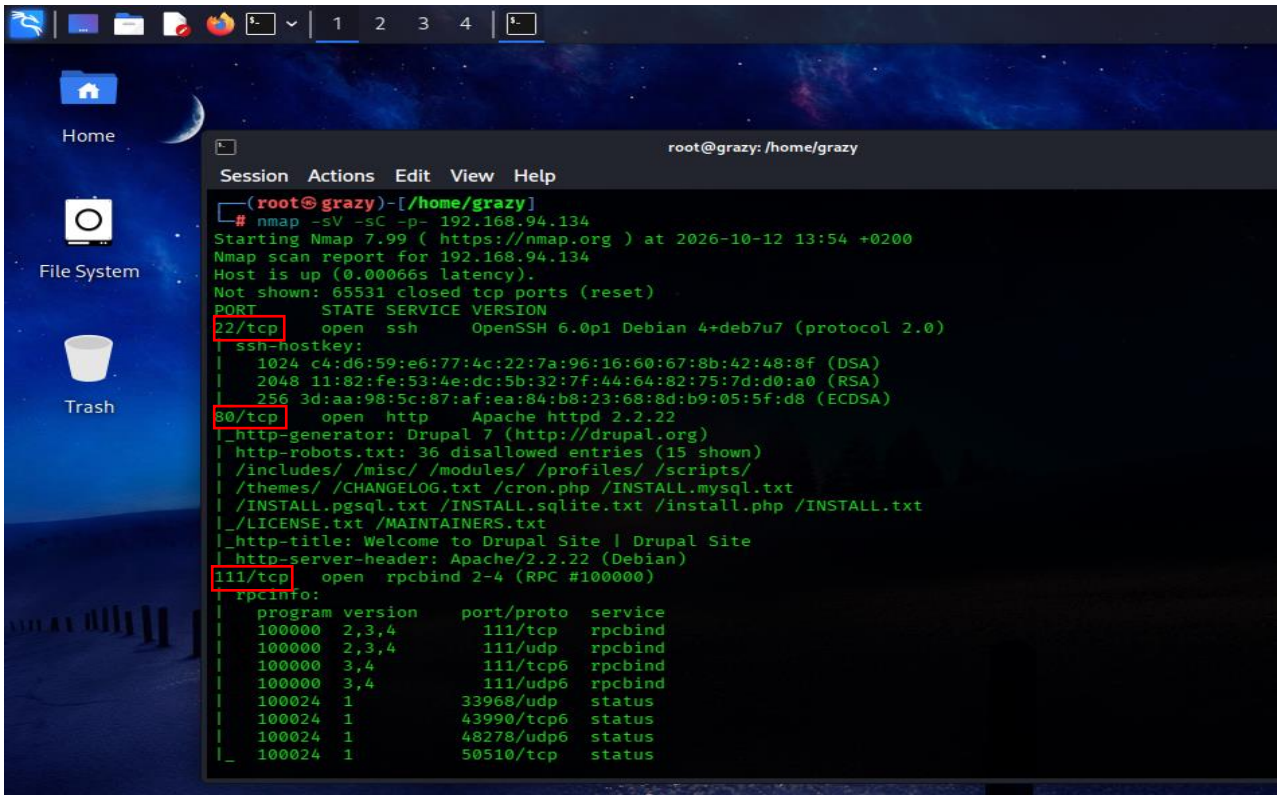
It provides evidence that the target machine was discovered on the local network before scanning it.

21. NMAP SCANNING

After identifying the DC-1 IP address, Nmap was used to identify open ports and services.

The scan identified the host as active and identified several open TCP ports.

The results included:



Nmap scan results for the DC-1 target

What this screenshot proves

This screenshot demonstrates that Nmap was used to identify the network services exposed by the target.

22. ANALYSIS OF NMAP RESULTS

The Nmap results showed that DC-1 had multiple network services exposed.

Port 22 - SSH

Port 22 was open and associated with SSH.

SSH is commonly used for secure remote administration.

Port 80 - HTTP

Port 80 was open and associated with HTTP.

The web server was identified as Apache running on Debian.

This service was selected for the next stage of the practical.

Port 111 - RPCbind

Port 111 was open and associated with RPCbind.

This demonstrated that a single system can expose multiple network services.

From a security perspective, every exposed service contributes to the system's attack surface and should therefore be understood and appropriately secured.

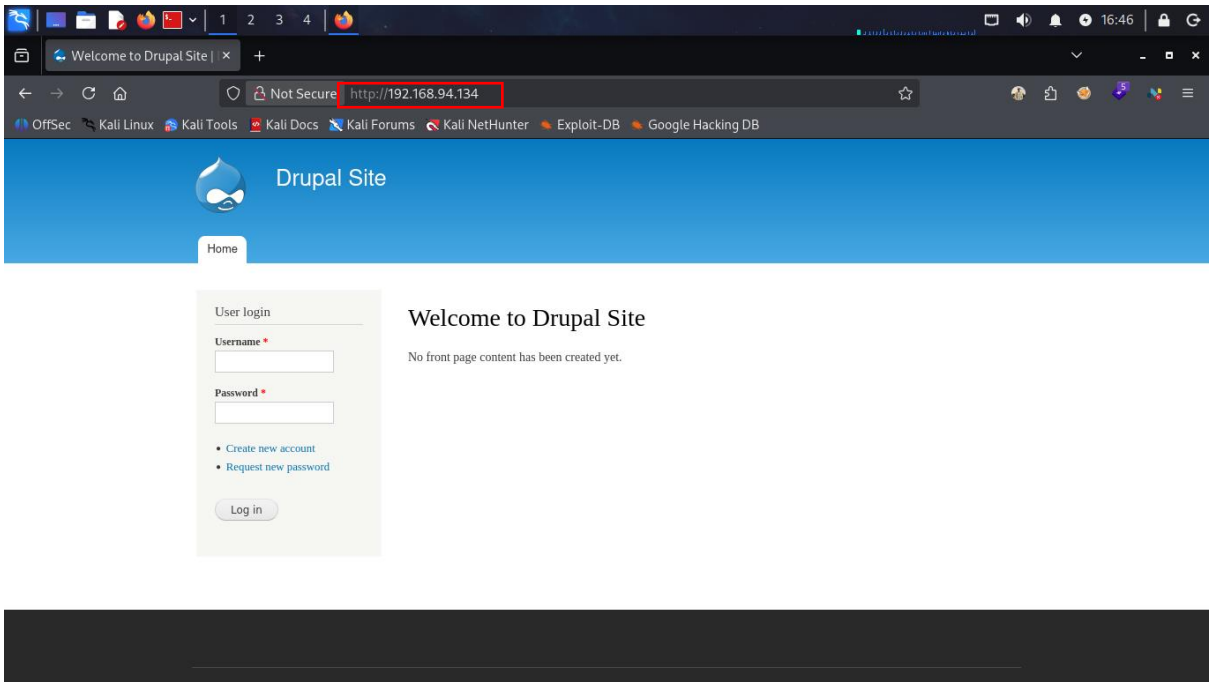
23. ACCESSING THE DC-1 WEB SERVER

After identifying HTTP on port 80, the following address was entered into Firefox: http://192.168.94.134

The browser successfully connected to the DC-1 web server.

Accessing the web server generated network communication between Kali Linux and DC-1.

This communication was then captured using Wireshark.



Accessing the DC-1 HTTP web server

What this screenshot proves

It proves that the HTTP service identified by Nmap was accessible and that accessing the service generated network traffic that could subsequently be captured with Wireshark.

24. WIRESHARK PACKET CAPTURE

Wireshark was opened on Kali Linux and the appropriate network interface was selected.

Packet capture was started before accessing the DC-1 web server.

When Firefox accessed: http://192.168.94.134 ,network traffic was generated between Kali Linux and DC-1.

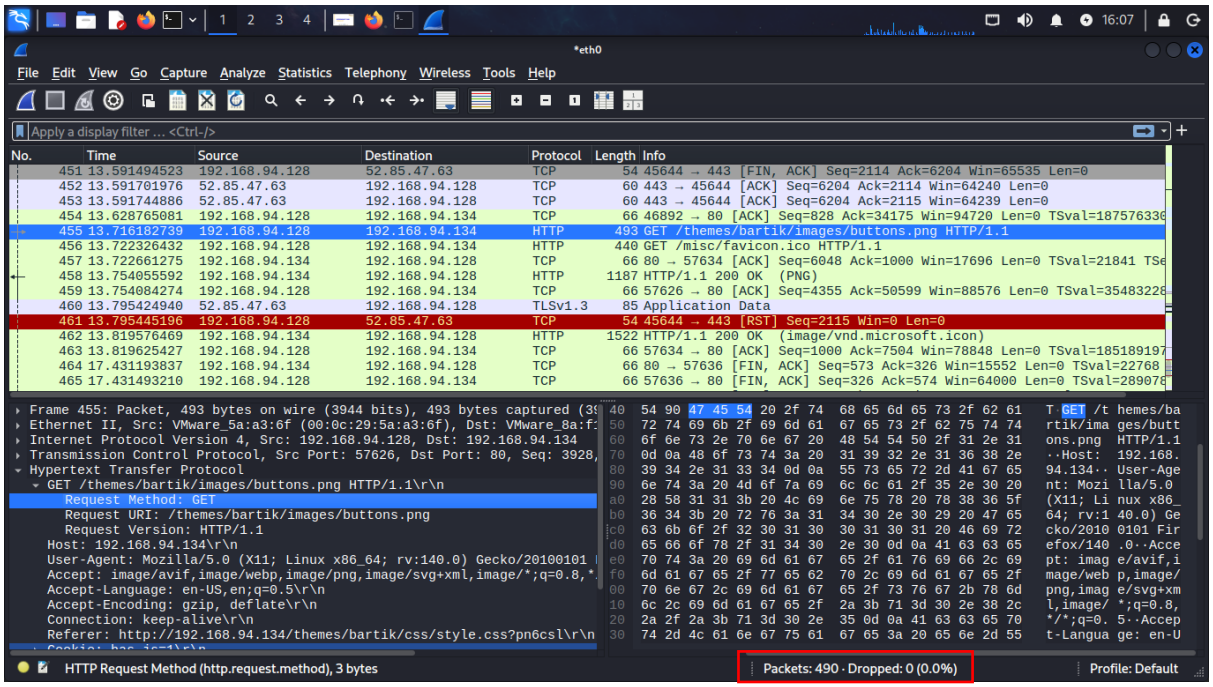
The Wireshark capture showed:

- Packets captured: 490
- Dropped: 0.0%

This means that 490 packets were captured during the practical and Wireshark reported no dropped packets.

The captured traffic included communication between:

- Source: 192.168.94.128
- Destination: 192.168.94.134



Wireshark packet capture of communication between Kali Linux and DC-1

What this screenshot proves

It provides evidence that network traffic was captured while accessing the target's HTTP service.

25. WIRESHARK HTTP PACKET ANALYSIS

After capturing the network traffic generated while accessing the DC-1 web server, a relevant HTTP packet was selected for detailed analysis.

The selected packet was Frame 455. The packet represented communication from the Kali Linux machine to the DC-1 web server.

The packet contained the following information:

1. Source IP: 192.168.94.128
2. Destination IP: 192.168.94.134
3. Protocol: HTTP
4. Source Port: 57626
5. Destination Port: 80

The packet also contained an HTTP request for a resource on the web server.

The request was:

GET /themes/bartik/images/buttons.png HTTP/1.1

This indicates that the browser was requesting an image file from the DC-1 web server.

It provides direct evidence that Wireshark captured and allowed detailed examination of HTTP communication between the Kali Linux machine and the DC-1 web server.

26. DETAILED ANALYSIS OF THE HTTP REQUEST

The selected HTTP packet provided information from both the transport and application layers of the network communication.

26.1 HTTP Request Method

The HTTP request used the **GET method**.

The GET method is used by a client, such as a web browser, to request a resource from a web server.

In this case, the browser requested an image file from the DC-1 web server.

26.2 Requested Resource

The requested resource was:

/themes/bartik/images/buttons.png

This indicates that the browser was requesting an image used by the webpage.

This also demonstrates that loading a webpage does not necessarily involve only one request. A browser may request HTML documents, images, stylesheets, scripts, and other resources separately.

26.3 HTTP Version

The request used:

HTTP/1.1

HTTP/1.1 is a version of the HTTP protocol used for communication between web clients and servers.

26.4 Host Information

The request identified the host as:

192.168.94.134

This corresponds to the IP address assigned to the DC-1 web server.

The Host information helps identify the server to which the HTTP request was directed.

26.5 User-Agent Information

The HTTP request also contained User-Agent information.

The User-Agent identifies information about the client making the request, including the browser and operating system.

In this practical, the captured information indicated that the request originated from Firefox running on Linux.

This demonstrates that HTTP headers can reveal information about the client communicating with a web server.

26.6 TCP Information

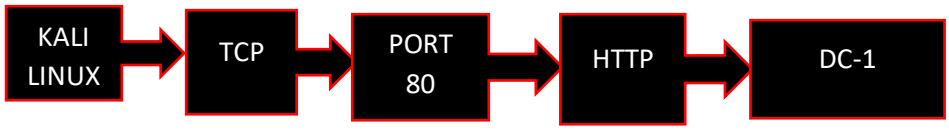
The HTTP request was transported using TCP.

The communication used:

- Source port: 57626
- Destination port: 80

The source port was a temporary client-side port, while port 80 was the standard port associated with HTTP.

The communication can therefore be summarized as:



This demonstrates the relationship between the different layers involved in network communication.

27. ANALYSIS OF THE PACKET USING THE OSI MODEL

The captured HTTP packet can be related directly to the OSI Model.

At the **Data Link Layer**, Ethernet communication is used to transfer frames across the local network.

At the **Network Layer**, IP addresses identify the source and destination systems:

- Source: 192.168.94.128
- Destination: 192.168.94.134

At the **Transport Layer**, TCP provides reliable communication between the two systems.

The packet used:

- Source port: 57626
- Destination port: 80

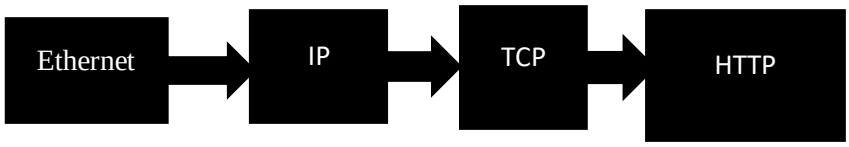
At the **Application Layer**, HTTP was used to request a resource from the web server.

The HTTP request was:

GET /themes/bartik/images/buttons.png HTTP/1.1

Therefore, the captured communication demonstrates how several networking layers work together to deliver an application request.

The relationship can be represented as:



This practical observation connects the theoretical OSI Model to actual network communication captured using Wireshark.

28. HTTP AND NETWORK SECURITY

The practical also demonstrated why understanding HTTP traffic is important in cybersecurity.

HTTP does not provide encryption for the application data being transmitted. As a result, information contained within HTTP requests and responses can potentially be observed when network traffic is captured.

In this practical, Wireshark allowed the HTTP request to be examined and revealed information including:

- Source IP address
- Destination IP address
- Source port
- Destination port
- HTTP method
- Requested resource
- HTTP version
- Host information
- User-Agent information

This demonstrates that network traffic can contain useful information about communication between a client and server.

From a cybersecurity perspective, this type of visibility can be useful during:

- Network monitoring
- Troubleshooting
- Security investigations
- Incident response
- Traffic analysis

It also demonstrates why organizations need appropriate security controls and why sensitive web communication should use encrypted protocols where appropriate.

29. RELATIONSHIP BETWEEN NMAP AND WIRESHARK

Nmap and Wireshark served different purposes during the practical.

Nmap was used to determine which services were available on the target, while Wireshark was used to examine the actual network communication generated by those services.

The Nmap scan identified HTTP on:

Port 80

The web service was then accessed through Firefox.

Wireshark subsequently captured traffic generated by that communication and showed HTTP traffic using:

Destination port: 80

This created a direct relationship between the two tools.

- Nmap answered the question: What network services are exposed by the target?
- Wireshark answered the question: What communication is actually taking place between the systems?

The practical therefore demonstrated that different cybersecurity tools can provide different levels of network visibility.

30. PRACTICAL FINDINGS

The practical produced several important findings.

Finding 1: The target was reachable

DC-1 was successfully identified on the local network at: 192.168.94.134

Finding 2: Multiple services were exposed

Nmap identified:

- SSH on port 22
- HTTP on port 80
- RPCbind on port 111

This showed that the target was running multiple network services.

Finding 3: HTTP was accessible

The HTTP service on port 80 was successfully accessed through Firefox.

Finding 4: Web access generated network traffic

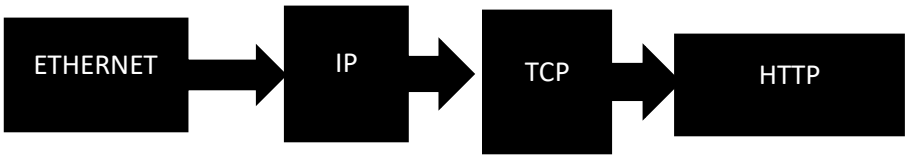
Accessing the DC-1 web server generated network traffic that was captured by Wireshark.

Finding 5: HTTP communication could be inspected

Frame 455 showed an HTTP GET request from Kali Linux to DC-1.

Finding 6: Multiple network layers were visible

The packet contained information associated with:



This demonstrated how multiple networking protocols work together during communication.

Finding 7: Webpages generate multiple requests

The captured request for:

/themes/bartik/images/buttons.png

demonstrated that a browser may request additional resources when loading a webpage.

31. SECURITY IMPORTANCE OF THE PRACTICAL

The practical demonstrated why network visibility is important in cybersecurity.

The Nmap scan provided information about the services exposed by the target system. This information can help security professionals understand the system's attack surface.

Wireshark provided a deeper level of visibility by allowing the actual communication to be inspected.

For example, the HTTP packet revealed:

- The communicating systems
- The protocol being used
- The ports involved
- The requested resource
- Client information
- Web server information

This type of visibility can assist security professionals in identifying unexpected communication, investigating incidents, troubleshooting network problems, and understanding how applications communicate.

The practical also demonstrated that knowing only which ports are open is not enough. A security analyst may also need to examine the traffic moving through those services to understand what is actually happening on the network.

32. KEY LESSONS LEARNED

The practical and theoretical work provided the following key lessons:

- Computer networks allow devices and systems to communicate.
- The OSI Model provides a structured way to understand network communication.
- The TCP/IP model describes protocols used by modern networks.
- IP addresses identify source and destination systems.
- Ports identify network services.
- TCP provides reliable and ordered communication.
- UDP provides communication with lower overhead.
- DNS translates domain names into IP addresses.
- DHCP provides network configuration information automatically.
- Routers connect different networks.
- Switches connect devices within local networks.
- Firewalls control network traffic according to security rules.
- VPNs can provide secure remote access to private resources.
- Nmap can identify open ports and services.
- Wireshark can capture and analyze network packets.

- HTTP communication can be inspected at the packet level.
- A single web request can involve multiple networking layers.
- Nmap and Wireshark provide complementary network visibility.
- Practical packet analysis helps connect theoretical networking concepts with real network communication.

33. CHALLENGES AND OBSERVATIONS

- ✓ One observation from the practical was that the DC-1 machine exposed multiple network services.

The Nmap scan identified:

- 22/tcp - SSH
- 80/tcp - HTTP
- 111/tcp - RPCbind

This demonstrated that a single system can provide multiple services simultaneously.

- ✓ Another observation was the large amount of traffic generated when accessing a web server.

Wireshark captured 490 packets, with 0.0% reported as dropped.

The selected HTTP packet showed:

GET /themes/bartik/images/buttons.png HTTP/1.1

This demonstrated that loading a webpage can generate multiple requests between a browser and a web server.

- ✓ A further observation was that the information contained within a packet can be used to understand what communication is taking place. By examining the IP addresses, ports, protocols, and HTTP information, it was possible to reconstruct important aspects of the communication between Kali Linux and DC-1.

34. CONCLUSION

Computer networking is an important foundation of cybersecurity because modern systems rely on networks to communicate, exchange information, and provide services.

This task provided an understanding of networking concepts including the OSI Model, TCP/IP Model, IP addressing, subnetting, DNS, DHCP, TCP, UDP, network devices, firewalls, VPNs, network packets, Nmap, and Wireshark.

The practical component was performed in a controlled virtual environment using Kali Linux and the intentionally vulnerable DC-1 machine from VulnHub.

The practical began with network discovery, where DC-1 was identified at 192.168.94.134. Nmap was then used to identify the services running on the target. The scan identified SSH on port 22, HTTP on port 80, and RPCbind on port 111.

The HTTP service running on port 80 was accessed using Firefox. Wireshark was then used to capture the resulting network traffic.

A total of 490 packets were captured, with 0.0% reported as dropped. Frame 455 was selected for detailed analysis and showed communication from Kali Linux at 192.168.94.128 to DC-1 at 192.168.94.134.

The communication used TCP source port 57626 and destination port 80. The captured HTTP request was:

GET /themes/bartik/images/buttons.png HTTP/1.1

The packet analysis demonstrated how Ethernet, IP, TCP, and HTTP work together during network communication.

The practical also demonstrated the complementary roles of Nmap and Wireshark. Nmap provided information about the services exposed by the target, while Wireshark provided detailed visibility into the actual traffic generated when one of those services was accessed.

Overall, the task provided practical experience in network discovery, service identification, packet capture, and packet analysis. It also demonstrated how theoretical networking concepts can be observed directly through real network traffic, providing a stronger foundation for further cybersecurity learning.